

Starting with Grid Engine

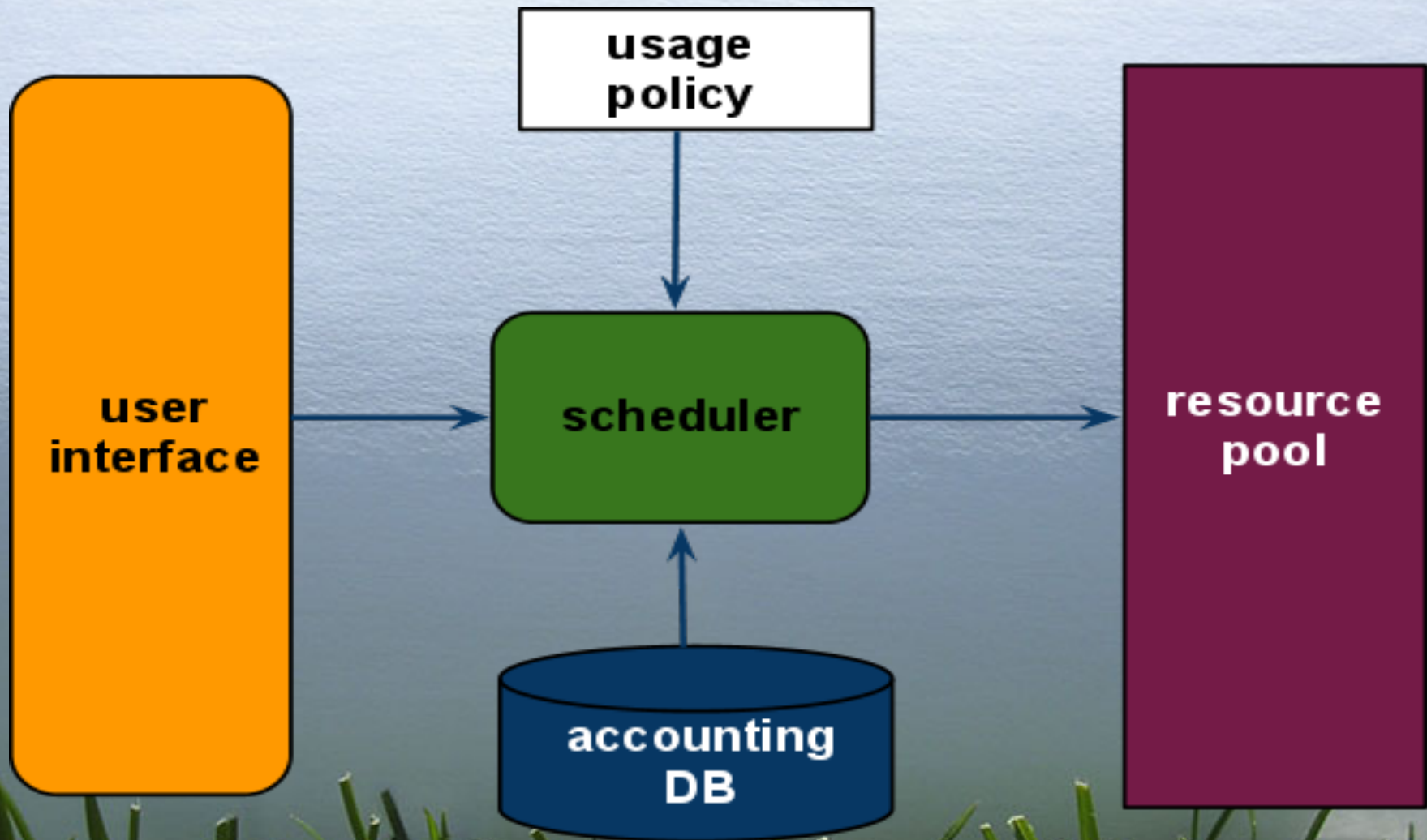
Job Submission and Scheduling at ACEnet

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Topics Covered

- what is a job scheduler?
- priority, queues
- N1GE commands
- options, directives
- some examples

What is a Job Scheduler?

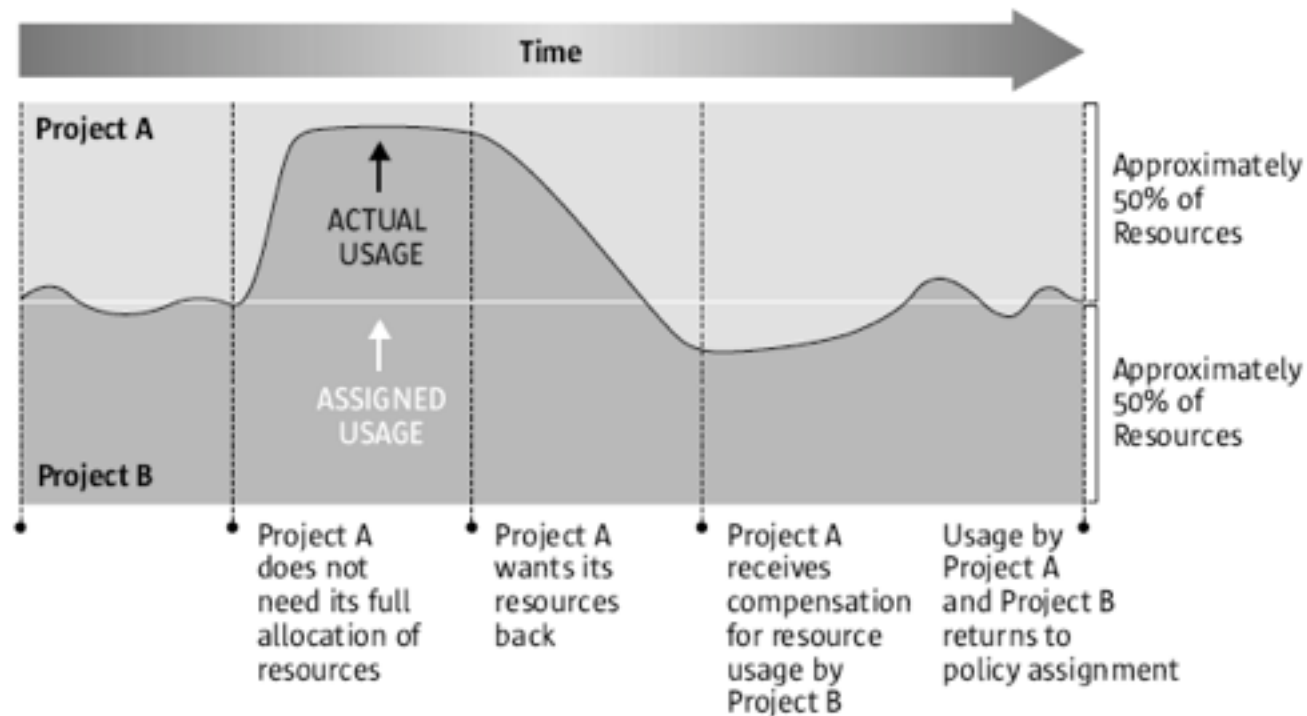


What is a Job Scheduler?

- ACEnet's job scheduler - N1 Grid Engine
- basic scheduling algorithm:
 - assign priorities to waiting jobs
 - don't interrupt running jobs
 - when resources are freed:
 - run highest priority job that can be satisfied by available resources
 - iterate until all resources are committed or all jobs have been checked
 - wait for new resources

Priority: "fairshare"

1. Share Tree Policy



<http://gridengine.info/2005/09/30/pretty-pictures-explain-functional-vs-sharetree-scheduling>

Queues

- ☐ what a queue is:
 - abstract container for running jobs
- what a queue is not:
 - waiting list
 - group of compute hosts
- queues have a finite size, e.g. long.q at mahone has 24 slots
- if the requested runtime is less than the maximum for a queue, the job can go there

Queues at ACEnet

- ☐ short.q - less than 48 hours
- medium.q - less than 2 weeks
- long.q - less than 1 month
- test.q - less than 1 hour (requires test=true flag)
- restricted queues:
 - Green ACEnet - demirov.q, cmms.q, tarasov.q
- special purpose queues:
 - gaussian.q, sub.q

N1GE Commands

- qsub submit an N1GE batch job
- qrsh start an interactive session
- qstat check status of running/queued jobs
- qdel delete job from list
- qalter alter run parameters of a job
- in-house scripts of interest:
 - qsum displays CPU usage by user and queue
 - showq shows active and waiting jobs

Submitting Jobs with qsub

- on the command line:

```
qsub -l h_rt=0:1:0 /usr/bin/hostname
```

- in a batch script:

```
qsub batch_file.sh
```


Job Monitoring with qstat

dshaw@mahone: N1GE_examples > qstat

job-ID	prior	name	user	state	submit/start at	queue	slots	ja-task-ID
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84296	0.00000	test	dshaw	qw	01/28/2010 11:14:50		1	
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...

dshaw@mahone: N1GE_examples > qstat

job-ID	prior	name	user	state	submit/start at	queue	slots	ja-task-ID
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84296	0.58654	test	dshaw	r	01/28/2010 11:14:55	test.q@cl064.smu.acenet.ca	1	
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Deleting Jobs with qdel

- □ general format of qdel:

qdel <Job ID>

- e.g. deleting job # 12345:

qdel 12345

Specifying Your Job Parameters

- ☐ jobs have two parts:
 - *options/directives*: info to N1GE about needed resources
 - *the tasks*: what you want N1GE to do on your behalf
- resource specification can be done on the command line as options to qsub or put in a batch file
- in the batch file, these options are called *directives*

Common Options/Directives

- ☐ hard run time:

- -l h_rt=hhh:mm:ss
- example: -l h_rt=1:0:0
- the only option that is ***required***

- shell:

- -S <path to shell>
- example: -S /bin/bash
- can specify shells that are in use at ACEnet
- default is bash

More Common Options/Directives

- ☐ job name:
 - -N <job_name>
 - example: -N thisismyjob
 - default: name of submit script
- stdout/stderr to file specification:
 - output/error streams go to a file
 - -o <output file>
 - -e <error file>
 - join together and put in one file, use -j y
 - example: -o my.out -j y

More Common Options/Directives

- start job in current working directory:
 - -cwd
 - default - start in home directory
- hard limit on virtual memory ***per core***:
 - -l h_vmem=n<K,M,G>
 - example: -l h_vmem=1G
 - must be integer units, decimals ignored
 - defaults set on each machine:
 - brasdor and glooscap - 500M
 - fundy, mahone and placentia - 2G
 - courtenay - no default set

●

More Common Options/Directives

- ☐ parallel jobs:

- `-pe <environment> <number of cores>`
- example (MPI): `-pe ompi 4`
- example (OpenMP): `-pe openmp 2`
- two parallel environments on all machines - ompi and openmp
- on placentia, special parallel environment for Gaussian jobs (gaussian)
- default is serial work so you must set this explicitly for parallel jobs

An Example Script

- a very simple script:

```
#$ -l h_rt=0:5:0  
date  
hostname  
sleep 30  
date
```


Another Example Script

```
#$ -cwd  
#$ -S /bin/tcsh  
#$ -N gauss_run1  
#$ -o gauss_run1.out -j y  
#$ -l h_vmem=5G  
#$ -l h_rt=24:0:0  
#$ -pe gaussian 4
```

date

source /usr/local/lib/g09.sh

g09 < mybigmp2job.com > mybigmp2job.log



try it yourself!

Resources

- ☐ ACEnet N1GE page:
 - https://wiki.ace-net.ca/index.php/Job_control
- N1GE 6.1 users guide:
 - <http://docs.sun.com/app/docs/doc/820-0699>
- ACEnet support:
 - support@ace-net.ca
 - your local CRC